



वश्निवा RACING



R.V. COLLEGE OF ENGINEERING
NEWSLETTER-SEPTEMBER-2016



NEWSLETTER - SEPTEMBER 2016

Dear Sponsors, supporters and well wishers,

Formula Hybrid 2016, indeed turned out to be the moment Ashwa longed to cherish. A lot to learn, a lot to venture and a lot to celebrate for.

Finishing 2nd in the design event, 2nd in the Project Management Presentation and securing the 4th position overall, left us all with a smile on our faces.

Our endeavor was fruitful only due to the faith and support that our sponsors have bestowed upon us. Our heartfelt gratitude to all our well-wishers as well!



The next challenge for Ashwa Racing is to earn a podium finish at Formula Hybrid, 2017. The manufacturing phase of RZ-X7H is currently ongoing, leaving us with 5 solid months for assembly and testing.

The support provided by RSST, our college, Sponsors and the faculty has been instrumental towards the project.

We would also like to thank the alumni who have been with us through our thick and thin.



CZECH REPUBLIC EVENT UPDATE

After bagging the 4th position overall at Formula Hybrid, the combustion team was looking to match this result at Czech Republic. ▢

With about 40 teams participating, it was a competitive event. The static events went well. Results are:

- 1) 12th in Business Plan (58/75).
- 2) Our best score in the design event (100/150).
- 3) And finally, 2nd position in the cost event.

This was our first podium finish during the event and things looked good at that point.

Our prototype was judged as the second most value for money car among all the other teams.

The car we took to the event was tested for over 300 km and was expected to win accolades in the dynamics events.

After clearing scrutiny (technical inspection, tilt test, noise and brake test) with ease, we were doing runs between the dynamic events. During this, our engine broke down, and we realised that we are the victims of sheer bad luck. The damage would take us at least a week to fix.

Overall we stood 21st out of 40 teams.

The team is now gearing up for the 2017 season. Valuable lessons learnt will be kept in mind and momentum in the static events will be maintained.

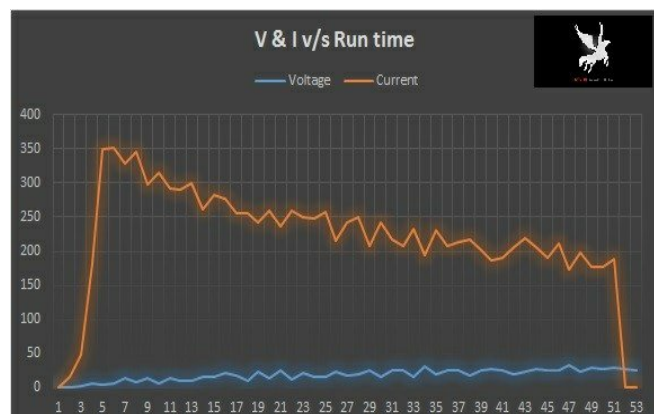




HYBRID TESTING

Testing of the X6-H has been done, and it gave us the needed thrust and confidence like never before. Some of the major validations conducted were:

In the in-house DAQ (Data Acquisition) system, the current and voltage of the traction motor were measured dynamically. The data logged in during the testing phase was used to validate the previous designs and also served as a reference for cross checking and validating the new powertrain designs.



Different motor controller modes were evaluated using timed acceleration runs and corresponding Voltage and Current values were recorded. This, in turn, resulted in an improved acceleration time by 5 seconds.

Pneumatic actuation of the clutch was tested using Festo components. The theoretical actuation force of 100N was practically validated to be 150N using the setup.

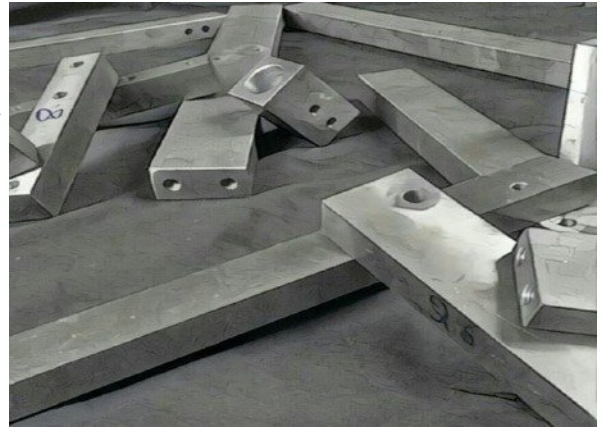
Also, improved designs of electronic circuits were tested in house and the PCB designs are being made accordingly.





RZ-X7H

The manufacturing of the new car has already begun in full swing, and we are all set for the upcoming Hybrid event.



After receiving the fixtures from VMS, the next few days were devoted towards fixtures setup on the baseplate.

Post this, the profiled tubes were received from Magod Laser. The consecutive days were solely devoted to TIG-welding of these tubes over the fixture setup. After four weeks of sleepless nights and endless working, the chassis was finally taken out of the base plate.



Also, on the combustion front, torsion test was carried out on the chassis to validate out our simulations and thus, helped us in providing much needed data for future prototypes.



ALUMNI FOCUS

As it is rightly said

“Entrepreneurs Create. Take risks. And Live their Passion”

None of these words are better suited to anyone else other than our founder, ArunRaj Subbaraj. Be it when he was just an engineering student at RVCE chasing his dreams, or now, when he is the Director of Lotus Paints, he is the reason why Ashwa Racing has been seeing the light of the day for the past 13 years.



For the members of Ashwa Racing, more than an institutor, he is a mentor. He has taught us to be adaptable. There were times when he told us to take the bull by the horns, instead of slowly steeping back. And then, there were times when he asked us to move even more slowly, making sure that all the team members are on the same page and that we all saw eye to eye as we moved together. There were other times when the same practices were important to continue, and then there were times to think out of the box for new innovation.

We take this opportunity to thank him for all his contributions and sacrifices because of which Ashwa Racing has reached where it stands today.

The entire team wishes you a belated Happy birthday and a great life ahead. We wish to take Ashwa Racing higher up the ladder and into the All-Star league in all listings with you and your guidance.



SHIPPING OF THE CAR



making material difference

TRIPADAM LOGISTICS PVT.LTD.

Tripadam-Logistics Pvt Limited is engaged in providing qualitative logistics services and packing services to their esteemed customers by combining lateral thought process, professional skills and personalized services.

The most important partnership, which we had been trying since years to materialize was a logistics partner. The team is grateful for all the support and encouragement given by Tripadam Logistics Pvt Ltd and we are proud to have them as our official logistic partner for the race season 2016-2017. Our car left the Chennai port on 18th of March and reached the event site on 21st of April, as per schedule.

The team conveys its heartfelt gratitude towards the company for extending its support and helping us overcome one of our biggest hurdles over the years: Logistics.

We wish to strengthen and take forward the partnership in the coming years.



OUR MANUFACTURING SPONSORS

We thank Mallar Group for their continued support in helping us with the milling of our baseplate, on which our chassis is built. We also thank them for helping us with our fixtures procurement. This marks the beginning of our tedious manufacturing phase. A special thank you to Mr Irfan Kazi for extending his support, when we needed it the most.



Middleby India is our latest in-kind sponsor on board, who has supported us with water jet machining for a large number of our aluminium components. We highly value the timely delivery of the machined components. Moreover, there is no taper on thick materials too, which is not easily achieved by other manufacturers.

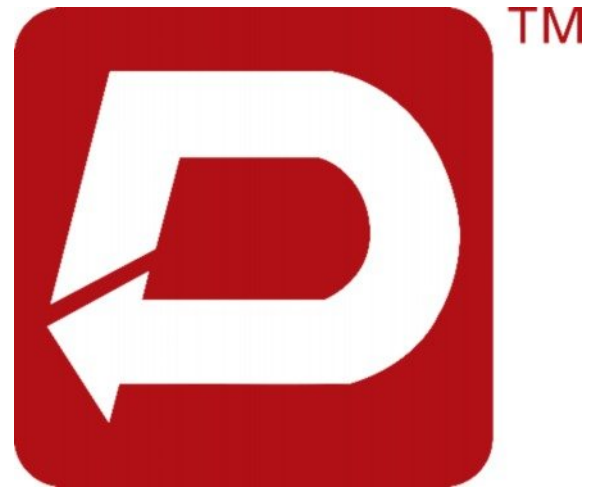
Magod Laser have been on board for the last 6 years, and we thank them for their continued support in the profiling of the tubes needed to fabricate the chassis. They also extend their help by laser cutting the sheet metal fixtures.





OUR MANUFACTURING SPONSORS

Dynamatics is the only manufacturer of Boeing and Airbus in India. We are thankful to Dynamatics for supporting us with manufacturing of the uprights and the pinion casing which are considered as the be one of the most difficult components to machine. We have been greatly appreciated in the vehicle dynamics designs only because of the manufacturing capabilities of Dynamatics.



We thank Agni Motors for their support in manufacturing of steering and suspension components by CNC machining. All the parts were precisely manufactured within the given tolerances.

Liebherr is one of the world's largest manufacturers of construction machinery. We thank them for manufacturing all of our sprockets and steering components like Rack and Pinion. All the manufacturing processes were done on a very short span of time and we are really grateful to them not just for manufacturing but also for providing material for the manufactured components.

LIEBHERR



OUR MANUFACTURING SPONSORS

DMG has extended its support to us for the ongoing race-season by manufacturing our hubs, which are yet another set of critical components in the wheel assembly. We are extremely grateful to DMG for their timely support.

The logo for DMG MORI, with 'DMG' in green and 'MORI' in red.The logo for Sansera Engineering, with 'SANSEARA' in blue and 'ENGINEERING' in grey.

Sansera Engineering, yet another long term sponsor of our team through financial backing, provided us with their manufacturing support as well for the ongoing race-season. We are extremely grateful to Sansera Engineering for their unending support. A thorough precision and quality inspection was done by them to ensure accuracy in the produced components.

Venus has supported us with milling and DRO of the fixture blocks of the combustion chassis. This is part of the fixture setup to initiate fabrication of the chassis. We thank Venus for their much needed and timely support.

The logo for VEI, with a stylized 'V' in grey and 'EI' in red.



OUR MANUFACTURING SPONSORS

Sun Precision has supported us with milling of the fixture blocks which allows us to dedicate funds towards other technical developments. We profoundly thank Mr Srinivas for working overtime and staying up late in the night, to get our job without affecting his schedule.

SUN PRECISION TOOLS

SHRI VENKATESHWARA HI-TECH

Since 2015, SV Hi-tech has helped us in manufacturing all the components that are part of the Drivetrain module. This includes the motor and generator mounts, Compound drive mount as well as the Differential mounts. We thank SV Hi-tech for their support and hope to continue this relationship for a long time to come.

Accu-Precision is a manufacturer of custom parts, sheet metal and assemblies for customers in aerospace and industrial sectors. We thank them for helping us out with drilling of the fixtures.





OUR MANUFACTURING SPONSORS

We are extremely grateful to Srinu, who have helped us with CNC milling of our steering and suspension components, namely: the rack casings, rocker shafts, casing mounts and C- Clamps.

**Srinu Engineering
Industries**

PRAGATI
Partnering Machine Tool Builders Worldwide

Pragati have been supporting us since 2015. They are into manufacturing of machine tool accessories. This season, they helped us with splining a few of our components like the tensioners and the differential mounts of the Drivetrain module.

MONTHLY AUDIT- JULY 2016



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ashwa RACING

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Ref. No: RVCE/ARF/FIN/2016

Accounts for the period from 01-07-2016 to 31-07-2016

RECIEPTS	AMOUNT
Opening Balance in Kotak Mahindra Bank RVCE Branch	
SB A/c 136010060793	₹ 1,122,144.93
SB A/c 136010058514	₹ 91,714.99
SB A/c 136010046665	₹ 209,733.62
FS Czech air ticket	₹ 380,000.00
Third year seed money	₹ 36,013.00
Refund from BMI	₹ 4,254.00
Sabic	₹ 50,000.00
BI-Technologies	₹ 5,000.00
Refund from mouser	₹ 555.23
refund for adhesive(intake manifold)	₹ 12,359.00
4th year seed money	₹ 18,007.00
Refund from MR aluminium	₹ 5,790.00
Cash in Hand	₹ 43,100.00
TOTAL	₹ 1,978,671.77

PAYMENTS BY	AMOUNT
Chassis & Suspension	₹ 58,665.00
Powertrain & Smp	₹ 1,900.00
Electrical	₹ 51,877.00
Brakes & Logistics	₹ 693,353.00
Miscellaneous	₹ 131,283.00
SMS & Wire Charges	₹ 775.18
Cash In Hand	₹ 43,400.00
Closing Balance in Kotak Mahindra Bank RVCE Branch	
SB A/c 136010060793	₹ 247,470.93
SB A/c 136010058514	₹ 190,589.81
SB A/c 136010046665	₹ 559,357.85
TOTAL	₹ 1,978,671.77

Checked and Found Correct



Abin P. Vincent
Finance Team
ashwa RACING

R.V. College of Engineering



MONTHLY AUDIT-AUGUST 2016

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Ref. No: RVCE/AR/FIN/2016

Accounts for the period from 01-08-2016 to 26-08-2016			
RECEIPTS		AMOUNT	
Opening Balance in Kotak Mahindra Bank RVCE Branch			
SB A/c 136010060793	₹	247,470.93	
SB A/c 136010058514	₹	190,589.81	
SB A/c 136010046665	₹	559,357.85	
4th year air ticket deposit	₹	20,000.00	
Third year seed money	₹	22,003.00	
Accommodation for 1st august for 10 people	₹	20,143.00	
refund from mouser	₹	50,000.00	
Cash In Hand	₹	43,400.00	
TOTAL	₹	1,152,964.59	
PAYMENTS BY		AMOUNT	
Chassis & Suspension	₹	6,138.00	
Powertrain & Smp	₹	2,874.00	
Electrical	₹	4,650.00	
Brakes & Logistics	₹	5,334.00	
Miscellaneous	₹	51,594.00	
SMS & Wire Charges	₹	14.38	
Cash In Hand	₹	54,345.00	
Closing Balance in Kotak Mahindra Bank RVCE Branch			
SB A/c 136010060793	₹	269,473.93	
SB A/c 136010058514	₹	212,003.18	
SB A/c 136010046665	₹	546,538.10	
TOTAL	₹	1,152,964.59	
Checked and Found Correct			
 Abin P. Vincent Finance Team ashwa racing, R.V. College of Engineering			



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KUMMAYA DESIGN STUDIOS





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FIÈ TECH
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LASER



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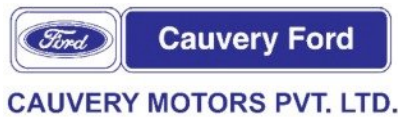


Suprajit

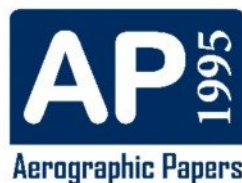
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PRAGATI
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MECKON

Securely Sealing Sentiments



SHRI VENKATESHWARA HI-TECH

SUN PRECISION TOOLS



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